

# The four–kernel row: $z$ and $y'$ integrated

Where the divergences are, and the terms separated

$$d^3 N_{\text{NLO}}/d^3 k \text{ with } \left[ \frac{\delta_{km}\delta_{ij}}{p^+ + k^+} - \frac{\delta_{im}\delta_{jk'}}{p^+} - \frac{\delta_{im}\delta_{jk'}}{k^+} \right]$$

## 0. The headline, before anything else

**This row has no ultraviolet divergence in the transverse integrations.** The whole transverse integrand is a product of *four* Weizsäcker–Williams kernels,

$$\frac{(y' - z)^m}{(y' - z)^2} \frac{(x' - y')^{k'}}{(x' - y')^2} \frac{(y - w)^i}{(y - w)^2} \frac{(x - z)^j}{(x - z)^2},$$

and the bracket that contracts them contains only  $\delta$ 's and longitudinal denominators — no transverse denominator at all. Each kernel is singular like  $1/|\text{sep}|$  at its own coincidence, and **no pair of points appears in two kernels**, so nowhere is there a  $1/\text{sep}^2$ . In two dimensions  $\int d^2\rho \rho^{-1} \sim \int d\rho$  converges, so every coincidence limit —  $y' \rightarrow z$ ,  $x' \rightarrow y'$ ,  $y \rightarrow w$ ,  $x \rightarrow z$ , and any pair of them at once — is integrable.

This is the structural difference from the row we did before. There the light–cone vertex supplied a *second*  $\frac{1}{(x-z)^2}$  on top of the kernel  $\frac{(z-x)^m}{(z-x)^2}$ ; contracting them gave  $\frac{1}{r^2}$ , and that is what produced the  $\log(1/m^2)$  and the  $P_{gg}$  coefficient. Here there is no such doubling.

What the  $z$  and  $y'$  integrals do contain is a *single* logarithm, coming from the opposite end — large  $|y' - z|$  — cut off by the oscillating phase. It is  $\log \frac{1}{Q^2 c^2}$  with  $Q$  the momentum conjugate to the integration variable, and it turns into the rapidity logarithms after the  $p^+$  integral. §5 separates it from the rest.

## 1. Notation, and the contraction

Write  $\xi = p^+/k^+$ , so that

$$\left[ \frac{\delta_{k'm}\delta_{ij}}{p^+ + k^+} - \frac{\delta_{jm}\delta_{ik'}}{p^+} - \frac{\delta_{im}\delta_{jk'}}{k^+} \right] = \frac{1}{k^+} \left[ \frac{\delta_{k'm}\delta_{ij}}{1 + \xi} - \frac{\delta_{jm}\delta_{ik'}}{\xi} - \delta_{im}\delta_{jk'} \right],$$

and abbreviate the four kernels

$$K_1^m = \frac{(y' - z)^m}{(y' - z)^2}, \quad K_2^{k'} = \frac{(x' - y')^{k'}}{(x' - y')^2}, \quad K_3^i = \frac{(y - w)^i}{(y - w)^2}, \quad K_4^j = \frac{(x - z)^j}{(x - z)^2}.$$

Contracting the four indices through the three  $\delta\delta$  structures collapses the whole transverse structure into three products of two *scalar* dipole kernels (checked symbolically, §8A):

$$K_1^m K_2^{k'} K_3^i K_4^j [\dots] = \frac{1}{k^+} \left[ \underbrace{\frac{(K_1 \cdot K_2)(K_3 \cdot K_4)}{1 + \xi}}_{T_1} - \underbrace{\frac{(K_1 \cdot K_4)(K_2 \cdot K_3)}{\xi}}_{T_2} - \underbrace{(K_1 \cdot K_3)(K_2 \cdot K_4)}_{T_3} \right].$$

The phase is  $e^{-ik \cdot (y' - w)} e^{-i\xi k \cdot (y' - z)}$ ; both factors depend on  $y'$ , only the second on  $z$ .

## 2. Both integrations reduce to one tensor master

In  $T_1, T_2, T_3$  the variable  $y'$  (or  $z$ ) appears only through  $K_1$  and one of  $K_2$  (or  $K_4$ ) — the other two kernels are spectators. Shifting to the separation as integration variable, each case becomes the *same* object:

$$B^{mj}(Q, c) \equiv \int d^2r e^{-iQ \cdot r} \frac{r^m}{r^2} \frac{(c - r)^j}{(c - r)^2}.$$

|                            | variable        | $Q$          | $c$      | spectators               |
|----------------------------|-----------------|--------------|----------|--------------------------|
| $\int d^2 y'$ at fixed $z$ | $r = y' - z$    | $(1 + \xi)k$ | $x' - z$ | $d = x - z, e = y - w$   |
| $\int d^2 z$ at fixed $y'$ | $\rho = z - y'$ | $-\xi k$     | $x - y'$ | $b = x' - y', e = y - w$ |

(In the first case the two phase factors combine,  $e^{-ik \cdot (y' - w)} e^{-i\xi k \cdot (y' - z)} = e^{-ik \cdot (z - w)} e^{-i(1+\xi)k \cdot r}$ ; in the second only the second factor depends on  $z$ ,  $e^{-i\xi k \cdot (y' - z)} = e^{+i\xi k \cdot \rho}$ .) Explicitly:

$$\int d^2 y' e^{-ik \cdot (y' - w)} e^{-i\xi k \cdot (y' - z)} [T_1 + T_2 + T_3] = \frac{e^{-ik \cdot (z - w)}}{k^+} \left[ \frac{(e \cdot d) \delta_{mj}}{(1+\xi) e^2 d^2} - \frac{d^m e^j}{\xi d^2 e^2} - \frac{e^m d^j}{e^2 d^2} \right] B^{mj}((1+\xi)k, x' - z),$$

$$\int d^2 z e^{-i\xi k \cdot (y' - z)} [T_1 + T_2 + T_3] = \frac{1}{k^+} \left[ -\frac{b^m e^j}{(1+\xi) b^2 e^2} + \frac{(b \cdot e) \delta_{mj}}{\xi b^2 e^2} + \frac{e^m b^j}{e^2 b^2} \right] B^{mj}(-\xi k, x - y').$$

As before, this presumes the Wilson lines  $U(y')$ ,  $U(z)$  are taken outside the transverse integral; the algebraic reduction above is pointwise and holds regardless.

### 3. The master, derived

Feynman-parametrise the two denominators:

$$\frac{1}{r^2(c-r)^2} = \int_0^1 \frac{d\alpha}{[\alpha r^2 + \bar{\alpha}(c-r)^2]^2} = \int_0^1 \frac{d\alpha}{[(r - \bar{\alpha}c)^2 + \Delta]^2}, \quad \Delta = \alpha \bar{\alpha} c^2.$$

Shift  $r = \bar{\alpha}c + u$ , so  $r^m = u^m + \bar{\alpha}c^m$ ,  $(c-r)^j = \alpha c^j - u^j$  and  $e^{-iQ \cdot r} = e^{-i\bar{\alpha}Q \cdot c} e^{-iQ \cdot u}$ . The three  $u$ -integrals are the two-dimensional Bessel transforms, with  $\lambda = |Q|\sqrt{\Delta}$ :

$$U_2 = \frac{\pi|Q|}{\sqrt{\Delta}} K_1(\lambda), \quad V_2^i = -i\pi Q^i K_0(\lambda), \quad T_2^{ij} = \pi \left[ \delta^{ij} K_0(\lambda) - \frac{Q^i Q^j \sqrt{\Delta}}{|Q|} K_1(\lambda) \right],$$

and expanding the numerator  $(u^m + \bar{\alpha}c^m)(\alpha c^j - u^j)$ :

$$B^{mj}(Q, c) = \int_0^1 d\alpha e^{-i\bar{\alpha}Q \cdot c} \left[ -T_2^{mj} + \alpha c^j V_2^m - \bar{\alpha} c^m V_2^j + \alpha \bar{\alpha} c^m c^j U_2 \right].$$

The  $\alpha$ -integral converges at both ends: at  $\alpha \rightarrow 0$  the pieces behave as  $\log \alpha$  (integrable) or tend to constants — e.g.  $\alpha \bar{\alpha} c^m c^j U_2 \rightarrow \pi c^m c^j / c^2$  — and likewise at  $\alpha \rightarrow 1$ . That convergence is the statement of §0: no regulator is needed.

The trace is an object already computed and checked against quadrature in the first note,

$$\delta_{mj} B^{mj}(Q, c) = -\overline{A_7(Q, c)}, \quad A_7(q, s) = \int d^2 r e^{iq \cdot r} \frac{r \cdot (r - s)}{r^2 (r - s)^2},$$

which agrees to  $10^{-15}$  (§8C) and fixes the normalisation of the whole construction.

### 4. Where the logarithm is

$B^{mj}$  is finite at  $r \rightarrow 0$  and  $r \rightarrow c$ ; the only large contribution comes from  $|r| \gg |c|$ , where the integrand tends to  $-e^{-iQ \cdot r} r^m r^j / r^4$  and the integral is held together only by the oscillation. Its size is therefore set by  $|Q|$ , and it grows logarithmically as  $|Q||c| \rightarrow 0$ . In the  $\alpha$ -representation this is visible as  $K_0(\lambda) \rightarrow \log \frac{2}{\lambda} - \gamma_E$  with  $\lambda = |Q||c|\sqrt{\alpha \bar{\alpha}}$ ; using  $\int_0^1 d\alpha \log \frac{1}{\sqrt{\alpha \bar{\alpha}}} = 1$ ,

$$B^{mj} \longrightarrow -\pi \delta^{mj} \left[ \log \frac{2}{|Q||c|} - \gamma_E + 1 \right] + \pi \frac{Q^m Q^j}{Q^2} + \pi \frac{c^m c^j}{c^2} + O(|Q||c|).$$

## 5. The separation

$$B^{mj}(Q, c) = \underbrace{-\frac{\pi}{2} \delta^{mj} \log \frac{4e^{-2\gamma_E}}{Q^2 c^2}}_{\text{(I) the logarithm}} \underbrace{-\pi \delta^{mj} + \pi \frac{Q^m Q^j}{Q^2} + \pi \frac{c^m c^j}{c^2}}_{\text{(II) finite, } Q\text{- and } c\text{-direction terms}} + \underbrace{\mathcal{B}^{mj}(Q, c)}_{\text{(III) the rest}}$$

with  $\mathcal{B}^{mj}$  defined as the difference — exactly, by the  $\alpha$ -integral of §3 minus (I)+(II) — and vanishing as  $O(|Q||c|)$ . The split is exact; it organises the answer by powers of  $|Q||c|$ , which is the produced-gluon transverse momentum times the transverse size of the dipole the kernel connects. Numerically (§8D),  $\mathcal{B}$  falls from 1.6 at  $|Q||c| \simeq 0.9$  to  $1.0 \times 10^{-2}$  at  $|Q||c| \simeq 10^{-3}$ , against a  $B$  of size  $\sim 24$  there.

Two remarks. Group (I) is a multiple of  $\delta^{mj}$ , so in the contractions of §2 it is picked up by *all three* terms: by  $T_2$  (or  $T_1$  in the  $z$ -integration) through the trace, giving  $-\pi \log \frac{4e^{-2\gamma_E}}{Q^2 c^2}$ , and by the other two through  $\delta_{mj} d^m e^j = d \cdot e$ . And group (I) has exactly the functional form of the ultraviolet logarithm of the previous row,  $\pi \log \frac{4e^{-2\gamma_E}}{q^2 m^2}$  — but here the short-distance scale is replaced by  $1/|c|$ , i.e. by the dipole size, because the divergence is at *large*  $|r|$ , not small. Nothing is regularised; nothing is subtracted.

## 6. The $p^+$ integral, and where the rapidity logarithms come from

With  $\xi = p^+/k^+$  the range  $p^+ \in [\Lambda, V - k^+]$  becomes  $\xi \in [\lambda, \Xi]$ ,  $\lambda = \Lambda/k^+$ ,  $\Xi = V/k^+ - 1$ , and the three weights are  $\frac{1}{1+\xi}$ ,  $-\frac{1}{\xi}$ ,  $-1$ . Because  $Q$  depends on  $\xi$ , so does the logarithm:

| integration   | $Q$          | logarithm (I)                                                               | behaviour at the endpoints                                                   |
|---------------|--------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| $\int d^2 y'$ | $(1 + \xi)k$ | $-\frac{\pi}{2} \delta^{mj} \log \frac{4e^{-2\gamma_E}}{(1+\xi)^2 k^2 c^2}$ | regular at $\xi \rightarrow 0$ ; $\sim -\pi \delta^{mj} \log \Xi$ at the top |
| $\int d^2 z$  | $-\xi k$     | $-\frac{\pi}{2} \delta^{mj} \log \frac{4e^{-2\gamma_E}}{\xi^2 k^2 c^2}$     | $\rightarrow +\pi \delta^{mj} \log \frac{1}{\xi}$ as $\xi \rightarrow 0$     |

In the  $z$ -integration the logarithm therefore hits the  $\frac{1}{\xi}$  pole of  $T_2$  and gives a **double** rapidity logarithm,

$$\int_{\lambda}^{\Xi} \frac{d\xi}{\xi} \log \frac{1}{\xi^2} = \log^2 \lambda - \log^2 \Xi = \ell_k^2 - \log^2 \Xi, \quad \ell_k = \log \frac{k^+}{\Lambda},$$

while the  $\xi$ -independent parts of groups (I) and (II) give the single logarithm  $\int_{\lambda}^{\Xi} \frac{d\xi}{\xi} = \ell_k + \log \Xi$ . The remaining weights are elementary,  $\int \frac{d\xi}{1+\xi} = \log \frac{1+\Xi}{1+\lambda} \rightarrow \log \frac{V}{k^+}$  and  $\int d\xi = \Xi - \lambda$ , the last being linear in the upper cutoff — a feature of the  $-\delta_{im} \delta_{jk'} / k^+$  term as written, worth checking against the source before it is used.

### Summary of the three groups, as asked.

**“UV” terms:** there are none. The transverse integrations are ultraviolet finite (§0), so no regulator, no  $1/\epsilon$ , no  $\overline{\text{MS}}$  subtraction is called for in this row.

**Logarithmic terms:** group (I),  $-\frac{\pi}{2} \delta^{mj} \log \frac{4e^{-2\gamma_E}}{Q^2 c^2}$ , from large  $|r|$  — these become the rapidity logarithms  $\ell_k$ ,  $\ell_k^2$  after the  $p^+$  integral.

**Finite terms:** group (II),  $-\pi \delta^{mj} + \pi \frac{Q^m Q^j}{Q^2} + \pi \frac{c^m c^j}{c^2}$ .

**Other terms:** group (III),  $\mathcal{B}^{mj}$ , exact as the one-parameter integral of §3 and  $O(|Q||c|)$ .

## 7. What to check before going further

- The upper limit  $\int_{\Lambda}^{V-k^+} dp^+$  together with the  $-\delta_{im} \delta_{jk'} / k^+$  term gives  $\int d\xi \cdot 1 = \Xi - \lambda$ , linearly divergent in  $V$ . Either  $V$  is a genuine physical cutoff here or that term pairs with something not shown.
- The colour structure carries  $U(y')$  and  $U(z)$ , so as written the transverse integrals of §2 are the ones with the Wilson lines taken outside. The reduction to  $B^{mj}$  (§1–§2) is pointwise and unaffected.
- If a  $\overline{\text{MS}}$  logarithm is expected from this row, it has to come from somewhere other than these four kernels — the power counting of §0 leaves no room for it.

## 8. Checks

```

A. the bracket collapses to three products of two scalar dipole kernels (sympy, exact)
   bracket == T1 + T2 + T3 : True

B. power counting. Each kernel is 1/|sep|; no pair of points appears in two
   kernels, so nowhere is there a 1/sep^2. In 2D  $\int d^2(\text{sep}) \sim \int \rho d\rho/\rho$ 
   converges, so the z and y' integrations are ULTRAVIOLET FINITE.
   (Contrast the earlier row, where the light-cone vertex supplied a second
    1/(x-z)^2 on top of the kernel (z-x)^m/(z-x)^2, giving 1/r^2 and a log.)

C. the tensor master  $B^{\{mj\}}(Q,c) = \int d^2r e^{-iQ.r} (r^m/r^2)((c-r)^j/(c-r)^2)$ 
   Q = [0.9 0.6]   c = [ 0.7 -0.4]
   alpha-representation vs direct 2D quadrature:
   B^00: rep +0.139713115-0.027594710j   quad +0.139815736-0.027636610j   rel 7.8e-04
   B^01: rep +0.405089716+1.555270518j   quad +0.406139542+1.554839032j   rel 7.1e-04
   B^10: rep -0.216624685-1.592494210j   quad -0.216683684-1.592470120j   rel 4.0e-05
   B^11: rep -2.593029830+0.512148819j   quad -2.593629623+0.512395339j   rel 2.5e-04
   (the quadrature is tail-limited here: the integrand falls only as 1/r^2 with a
    nonzero angular average. The trace gives an exact check instead:)
   trace B = -2.453316715316+0.484554108681j
   -conj(A7)= -2.453316715316+0.484554108681j   rel 1.2e-15
   (A7 is the master already verified against quadrature in the first note)

D. the exact separation
   B^{\{mj\}} = -(pi/2) d^{\{mj\}} log(4 e^{-2gamma}/(Q^2 c^2))      <- the logarithm
               - pi d^{\{mj\}} + pi Q^m Q^j/Q^2 + pi c^m c^j/c^2    <- constants
               + Bcal^{\{mj\}}(Q,c)                                  <- remainder, 0(|Q||c|)

   |Q||c|      max |Bcal^{\{mj\}}|      max |B^{\{mj\}}|
   0.87207      1.623e+00      2.64312
   0.26162      9.116e-01      6.04384
   0.08721      4.376e-01      9.43958
   0.02616      1.755e-01      13.21322
   0.00872      7.196e-02      16.66362
   0.00262      2.601e-02      20.44587
   0.00087      1.002e-02      23.89725

   trace of the log+constant part = -pi log(4 e^{-2gamma}/(Q^2 c^2)) :
   |Q||c|= 0.08721: trace -16.073013+0.313463j   predicted -16.056091
   |Q||c|= 0.00872: trace -30.523911+0.059522j   predicted -30.523659
   |Q||c|= 0.00087: trace -44.991232+0.008773j   predicted -44.991228

```

zint/src/nlo2\_setup.py — the symbolic contraction and the power counting. zint/src/nlo2\_master.py —  $B^{mj}$  and its quadrature check. zint/src/run\_nlo2.py — every number above, written to zint/RESULTS\_nlo2.txt.